Variable Selection for Wasserstein Robust Hypothesis Testing

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \le \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \le d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Convex formulation

For $S \subseteq [D]$, denote $V(S) = \min_{f \in \mathcal{F}_S} R(f)$ and define

$$\mathcal{X} = \{x_i^{(0)} : i \in [n_0]\} \cup \{x_j^{(1)} : j \in [n_1]\}.$$

Assume $n_0, n_1 \ge 1$, $\rho_0, \rho_1 > 0$, and that ℓ is finite on $\mathbb{R}$. Following the transport formulation in [3], we formulate $V(S)$ as the following finite-dimensional optimization problem. For each target label $y \in \mathcal{X}$, let $g_y \in \mathbb{R}$ denote its detector score.

Lemma 2.1 (Least favorable distributions)**.**

$$\begin{aligned}
V(S) = \max_{p, \gamma \ge 0} \quad & \sum_{y \in \mathcal{X}} \inf_{g_y \in \mathbb{R}} \{p_0(y)\ell(-g_y) + p_1(y)\ell(g_y)\} \\
\textit{subject to} \quad & \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = \frac{1}{n_k}, && i \in [n_k],\ k = 0, 1, \\
& p_k(y) = \sum_{i=1}^{n_k} \gamma_{i,y}^{(k)}, && y \in \mathcal{X},\ k = 0, 1, \\
& \sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \le \rho_k, && k = 0, 1.
\end{aligned} \tag{3}$$

Here $\gamma_{i,y}^{(k)}$ is the mass transported from $x_i^{(k)}[S]$ to $y[S]$, and $p_k(y)$ is the mass of class k at that target. The lemma obtains the least favorable target masses by maximizing their minimum weighted loss subject to the transport budgets.

Write $g = (g_y)_{y\in\mathcal{X}} \in \mathbb{R}^{\mathcal{X}}$, and let Γ_S denote the feasible transports in Eq. (3), after eliminating p. Sion's minimax theorem [2] allows the interchange

$$\begin{aligned} V(S) &= \max_{\gamma\in\Gamma_S} \inf_{g} \sum_{k=0}^{1} \sum_{i,y} \gamma_{i,y}^{(k)} \ell((-1)^{k+1} g_y) \\ &= \inf_{g} \max_{\gamma\in\Gamma_S} \sum_{k=0}^{1} \sum_{i,y} \gamma_{i,y}^{(k)} \ell((-1)^{k+1} g_y). \end{aligned}$$

Using Lagrangian duality for each class, write

$$\begin{aligned} \mathcal{L}_k = \sum_{i,y} \gamma_{i,y}^{(k)} \ell((-1)^{k+1} g_y) + \sum_i a_i^{(k)} \left(\frac{1}{n_k} - \sum_y \gamma_{i,y}^{(k)} \right) \\ + \lambda_k \left(\rho_k - \sum_{i,y} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \right). \end{aligned}$$

Here $a_i^{(k)} \in \mathbb{R}$ and $\lambda_k \geq 0$ are the multipliers for source-mass conservation and the transport budget, respectively. Taking the supremum over $\gamma^{(k)} \geq 0$ and minimizing the dual objective over g, a, λ gives

$$\begin{aligned} V(S) = \min_{g,a,\lambda} \quad & \frac{1}{n_0} \sum_i a_i^{(0)} + \frac{1}{n_1} \sum_j a_j^{(1)} + \rho_0\lambda_0 + \rho_1\lambda_1 \\ \text{subject to} \quad & \ell(-g_y) \leq a_i^{(0)} + \lambda_0 \|x_i^{(0)}[S] - y[S]\|_1, \qquad i \in [n_0],\ y \in \mathcal{X}, \\ & \ell(g_y) \leq a_j^{(1)} + \lambda_1 \|x_j^{(1)}[S] - y[S]\|_1, \qquad j \in [n_1],\ y \in \mathcal{X}, \\ & g \in \mathbb{R}^{\mathcal{X}}, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda_k \geq 0, \qquad k = 0, 1. \end{aligned} \tag{4}$$

Eq. (4) is a finite convex program for the selected coordinates. Appendix A proves the equivalence and recovers a detector from its solution.

To optimize the coordinate selection, define $z = \mathbf{1}_S$ and $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \leq d\}$. For $\gamma = (\gamma^{(0)}, \gamma^{(1)})$ with $\gamma^{(k)} \in \mathbb{R}_+^{n_k \times |\mathcal{X}|}$, define the minimum weighted loss over the finite scores in Eq. (4):

$$H(\gamma) = \inf_{g\in\mathbb{R}^{\mathcal{X}}} \sum_{y\in\mathcal{X}} \left[\sum_{i=1}^{n_0} \gamma_{i,y}^{(0)} \ell(-g_y) + \sum_{j=1}^{n_1} \gamma_{j,y}^{(1)} \ell(g_y) \right].$$

Define the average of the two class masses and the transport cost in coordinate m, respectively, by

$$\begin{aligned} M(\gamma) &= \frac{1}{2} \sum_{k=0}^{1} \sum_{i=1}^{n_k} \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)}, \\ c_k(\gamma)[m] &= \sum_{i=1}^{n_k} \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} |x_i^{(k)}[m] - y[m]|, \qquad k = 0, 1,\ m \in [D]. \end{aligned}$$

For a scalar $t \in \mathbb{R}$, write $t_+ = \max\{t, 0\}$. Following [1], the next result expresses the selection objective as a maximum of functions affine in z.

Proposition 2.2 (Convex selection function)**.** *For $z \in [0,1]^D$, define*

$$\begin{aligned} F(z) = \max_{\gamma \geq 0,\, \mu \in \mathbb{R}_+^D} \quad & 2\sqrt{H(\gamma)} - M(\gamma) - \mu^\top z \\ \textit{subject to} \quad & \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = \frac{M(\gamma)}{n_k}, \qquad i \in [n_k],\ k = 0, 1, \\ & \sum_{m=1}^{D} \big(c_k(\gamma)[m] - \rho_k \mu_m\big)_+ \leq \rho_k M(\gamma), \quad k = 0, 1. \end{aligned} \tag{5}$$

Then F is convex, and

$$F(\mathbf{1}_S) = V(S), \qquad \min_{|S| \leq d} V(S) = \min_{z \in \mathcal{Z}_d} F(z). \tag{6}$$

The row constraints make each class's total mass equal to $M(\gamma)$; this common mass is allowed to vary, whereas each class has unit mass in Eq. (3). Appendix B derives Eq. (5) from the epigraph of Eq. (4) by duality and a change of scale. Minimizing F over $\mathcal{Z}_d$ therefore solves the original problem; replacing $\mathcal{Z}_d$ by its continuous relaxation gives a lower bound.

3 An outer-approximation approach

We use an outer-approximation approach following [1], which iteratively refines affine lower bounds on F over $\mathcal{Z}_d$. Since H is nonnegative and concave, Eq. (5) is a convex optimization problem for fixed z. For an optimizer (γ^t, μ^t) at z^t, define

$$\begin{aligned} c_t(z) &= 2\sqrt{H(\gamma^t)} - M(\gamma^t) - (\mu^t)^\top z \\ &= F(z^t) - (\mu^t)^\top (z - z^t). \end{aligned} \tag{7}$$

The feasible set in Eq. (5) is independent of z, so $c_t(z) \leq F(z)$ for every $z \in [0,1]^D$, with equality at z^t. After t evaluations, we minimize the accumulated lower bounds to obtain the next selection:

$$\begin{aligned} (z^{t+1}, L_t) \in \ \operatorname*{arg\,min}_{z \in \mathcal{Z}_d,\, \eta \geq 0} \quad & \eta \\ \text{subject to} \quad & \eta \geq c_s(z), \qquad s = 1, \ldots, t. \end{aligned} \tag{8}$$

The value L_t is a lower bound on $\mathrm{OPT} = \min_{z \in \mathcal{Z}_d} F(z)$, and the best evaluated value $U_t = \min_{s \leq t} F(z^s)$ is an upper bound.

Algorithm 1 alternates between evaluating a selection and solving Eq. (8). It retains the best evaluated detector and stops when $U_t - L_t \leq \varepsilon$. Since adding coordinates cannot increase $V(S)$, the budget can also be imposed as $\mathbf{1}^\top z = d$ when $d \leq D$.

When a solver stops early, feasible solutions of Eq. (4) give upper bounds, and feasible solutions of Eq. (5) give cuts $2\sqrt{H(\gamma)} - M(\gamma) - \mu^\top z$. Use the solver's global lower bound for Eq. (8) as L_t.

Algorithm 1 Outer approximation for variable selection

Require: Samples $x_i^{(k)}$, radii $\rho_k > 0$, budget d, tolerance $\varepsilon > 0$, initial $z^1 \in \mathcal{Z}_d$

$U \leftarrow +\infty$, $t \leftarrow 1$
loop
  $S^t \leftarrow \{m : z_m^t = 1\}$
  Solve Eq. (4) at S^t for (g^t, a^t, λ^t) and value $v_t = V(S^t)$
  Solve Eq. (5) at z^t for (γ^t, μ^t); form c_t by Eq. (7)
  if $v_t < U$ **then**
    $U \leftarrow v_t$; store $S^\star \leftarrow S^t$ and $(g^\star, a^\star, \lambda^\star) \leftarrow (g^t, a^t, \lambda^t)$
  end if
  Solve Eq. (8) with $c_1, \ldots, c_t$ for (z^{t+1}, L_t)
  if $U - L_t \le \varepsilon$ **then**
    Recover $g^\star$ on $\mathbb{R}^{|S^\star|}$ by Eq. (9) in Appendix A
    return $S^\star$ and $f^\star(x) = g^\star(x[S^\star])$
  end if
  $t \leftarrow t + 1$
end loop

A Proof of Lemma 2.1

Fix $S \subseteq [D]$, and let v_S be the infimum in Eq. (4). We prove

$$\max_{\gamma \in \Gamma_S} H(\gamma) = v_S = V(S).$$

For $g \in \mathbb{R}^{\mathcal{X}}$, define

$$\Phi(\gamma, g) = \sum_{y \in \mathcal{X}} \left[\sum_{i=1}^{n_0} \gamma_{i,y}^{(0)} \ell(-g_y) + \sum_{j=1}^{n_1} \gamma_{j,y}^{(1)} \ell(g_y) \right].$$

Then

$$\begin{aligned} \max_{\gamma \in \Gamma_S} H(\gamma) &= \max_{\gamma \in \Gamma_S} \inf_g \Phi(\gamma, g) \\ &= \inf_g \max_{\gamma \in \Gamma_S} \Phi(\gamma, g) = v_S. \end{aligned}$$

The interchange follows from Sion's theorem [2]: Γ_S is compact and convex, and Φ is continuous, affine in γ, and convex in g. For fixed g, the maximizations separate by class and are feasible, bounded linear programs. Taking their Lagrangian duals gives Eq. (4), which yields the last equality.

We next establish $v_S \le V(S)$. For $\gamma \in \Gamma_S$ and each i, y, k, define $\widetilde{y}_i^{(k)}$ by $\widetilde{y}_i^{(k)}[S] = y[S]$ and $\widetilde{y}_i^{(k)}[[D] \setminus S] = x_i^{(k)}[[D] \setminus S]$, and set $\mathbb{P}_k^\gamma = \sum_{i,y} \gamma_{i,y}^{(k)} \delta_{\widetilde{y}_i^{(k)}}$. Then

$$\mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k^\gamma) \le \sum_{i,y} \gamma_{i,y}^{(k)} \|x_i^{(k)} - \widetilde{y}_i^{(k)}\|_1 = \sum_{i,y} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \le \rho_k.$$

For any $f(x) = g(x[S]) \in \mathcal{F}_S$, it follows that

$$R(f) \ge \Phi\big(\gamma, (g(y[S]))_{y \in \mathcal{X}}\big) \ge \inf_{u \in \mathbb{R}^{\mathcal{X}}} \Phi(\gamma, u).$$

Taking the maximum over Γ_S and the infimum over $\mathcal{F}_S$ gives $v_S \le V(S)$.

It remains to prove $V(S) \le v_S$. Let (g, a, λ) be any feasible solution of Eq. (4). Coincident projected targets have identical constraints, so their scores may be chosen equal. Set $g(y[S]) = g_y$. For $\xi \in \mathbb{R}^{|S|}$, define

$$E_k(\xi) = \min_{r \in [n_k]} \{a_r^{(k)} + \lambda_k \|x_r^{(k)}[S] - \xi\|_1\}, \qquad k = 0, 1.$$

For $\xi \notin \{y[S] : y \in \mathcal{X}\}$, choose the smallest minimizing indices i, j for $E_0(\xi), E_1(\xi)$ and write

$$r_0 = \|x_i^{(0)}[S] - \xi\|_1, \quad r_1 = \|x_j^{(1)}[S] - \xi\|_1, \quad b = \|x_i^{(0)}[S] - x_j^{(1)}[S]\|_1.$$

Define

$$\tau = \begin{cases} \min\{1, r_0/b\}, & b > 0, \\ 0, & b = 0, \end{cases} \qquad g(\xi) = (1-\tau) g_{x_i^{(0)}} + \tau g_{x_j^{(1)}}. \tag{9}$$

The triangle inequality implies $\tau b \le r_0$ and $(1-\tau)b \le r_1$. By convexity and Eq. (4),

$$\begin{aligned} \ell(-g(\xi)) &\le (1-\tau)\ell(-g_{x_i^{(0)}}) + \tau \ell(-g_{x_j^{(1)}}) \\ &\le a_i^{(0)} + \lambda_0 \tau b \le a_i^{(0)} + \lambda_0 r_0 = E_0(\xi), \\ \ell(g(\xi)) &\le (1-\tau)\ell(g_{x_i^{(0)}}) + \tau \ell(g_{x_j^{(1)}}) \\ &\le a_j^{(1)} + \lambda_1 (1-\tau) b \le a_j^{(1)} + \lambda_1 r_1 = E_1(\xi). \end{aligned} \tag{10}$$

These inequalities also hold at projected targets by Eq. (4). The finite minimizing-index rule makes g Borel measurable, and $|g| \le \max_y |g_y|$. For $\mathbb{P}_k \in \mathcal{P}_k$, let $\pi_i^{(k)}$ be the components of an optimal coupling from $\widehat{\mathbb{P}}_k$ to $\mathbb{P}_k$, with mass $1/n_k$ and $\sum_i \pi_i^{(k)} = \mathbb{P}_k$. For $k = 0, 1$,

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_k} \ell((-1)^{k+1} g(X[S])) &= \sum_i \int \ell((-1)^{k+1} g(x[S]))\, d\pi_i^{(k)}(x) \\ &\le \sum_i \int \left[a_i^{(k)} + \lambda_k \| x_i^{(k)}[S] - x[S] \|_1 \right] d\pi_i^{(k)}(x) \\ &\le \frac{1}{n_k} \sum_i a_i^{(k)} + \rho_k \lambda_k. \end{aligned}$$

Thus, for $f(x) = g(x[S])$,

$$v_S \le V(S) \le R(f) \le \sum_{k=0}^{1} \left(\frac{1}{n_k} \sum_i a_i^{(k)} + \rho_k \lambda_k \right).$$

Taking the infimum over feasible (g, a, λ) gives $V(S) \le v_S$, and hence $V(S) = v_S$.

To establish attainment, observe that finite scores admit feasible source bounds with $\lambda = 0$. On any finite objective sublevel B, the diagonal constraints give

$$0 \le a_i^{(k)} \le n_k B, \qquad 0 \le \lambda_k \le B/\rho_k.$$

The remaining constraints bound both $\ell(g_y)$ and $\ell(-g_y)$. Since convexity and the assumptions on ℓ imply $\ell(t) \to \infty$ as $t \to \infty$, every such sublevel is bounded and closed. Thus Eq. (4) attains its minimum, and its extension above attains $R(f) = V(S)$. Upper semicontinuity of H on Γ_S gives attainment in Eq. (3). □

B Proof of Proposition 2.2

Let $\mathcal{D}$ denote the feasible set in Eq. (5). Introduce an epigraph variable η in Eq. (4) and write $\|x_i^{(k)}[S] - y[S]\|_1 = \sum_m z_m |x_i^{(k)}[m] - y[m]|$ for $z = \mathbf{1}_S$. The diagonal constraints give $a_i^{(k)} \ge 0$, so the epigraph inequality implies $\sum_k \rho_k \lambda_k \le \eta$. Introducing w_{km} for $\lambda_k z_m$, consider the following

problem for fixed $z \in [0,1]^D$:

$$
\begin{aligned}
v(z) = \min_{g,a,\lambda,w,\eta} \quad & \eta \\
\text{subject to} \quad & \ell((-1)^{k+1} g_y) \le a_i^{(k)} + \sum_m w_{km} |x_i^{(k)}[m] - y[m]|, \quad k = 0,1,\ i \in [n_k],\ y \in \mathcal{X}, \\
& 0 \le w_{km} \le \lambda_k, \qquad\qquad k = 0,1,\ m \in [D], \\
& \sum_{k=0}^{1} \rho_k w_{km} \le \eta z_m, \qquad\qquad m \in [D], \\
& \sum_{k=0}^{1} \left(\frac{1}{n_k} \sum_i a_i^{(k)} + \rho_k \lambda_k \right) \le \eta, \\
& g \in \mathbb{R}^{\mathcal{X}}, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \eta \in \mathbb{R}.
\end{aligned}
\tag{11}
$$

We first show that, at $z = \mathbf{1}_S$, its projection onto (g, a, λ, η) is exactly the epigraph of Eq. (4). For any feasible tuple in Eq. (11), the coordinate constraint and $\rho_k > 0$ imply $w_{km} = 0$ for $m \notin S$. Consequently,

$$
\begin{aligned}
\ell((-1)^{k+1} g_y) &\le a_i^{(k)} + \sum_{m \in S} w_{km} |x_i^{(k)}[m] - y[m]| \\
&\le a_i^{(k)} + \lambda_k \|x_i^{(k)}[S] - y[S]\|_1,
\end{aligned}
$$

which gives one inclusion. Conversely, let (g, a, λ, η) belong to the epigraph of Eq. (4) and put $w_{km} = \lambda_k z_m$. The loss constraints are unchanged and $0 \le w_{km} \le \lambda_k$. Moreover,

$$
\begin{aligned}
\sum_k \rho_k w_{km} &= z_m \sum_k \rho_k \lambda_k \\
&\le z_m \sum_k \left(\frac{1}{n_k} \sum_i a_i^{(k)} + \rho_k \lambda_k \right) \le z_m \eta,
\end{aligned}
$$

where the first inequality follows from $a_i^{(k)} \ge \ell((-1)^{k+1} g_{x_i^{(k)}}) \ge 0$. This proves the reverse inclusion and hence $v(\mathbf{1}_S) = V(S)$.

Using nonnegative Lagrange multipliers $\gamma, \beta, \mu, \alpha$ for the loss, weight, coordinate, and epigraph inequalities, respectively, we obtain

$$
\begin{aligned}
\mathcal{L} = (1 - \alpha - \mu^\top z)\eta &+ \sum_{k,i,y} \gamma_{i,y}^{(k)} \ell((-1)^{k+1} g_y) \\
&+ \sum_{k,i} a_i^{(k)} \left(\frac{\alpha}{n_k} - \sum_y \gamma_{i,y}^{(k)} \right) + \sum_k \lambda_k \left(\rho_k \alpha - \sum_m \beta_{km} \right) \\
&+ \sum_{k,m} w_{km} \left[\beta_{km} + \rho_k \mu_m - c_k(\gamma)[m] \right].
\end{aligned}
$$

Minimization over g gives $H(\gamma)$. Minimization over the free variables a, η and the nonnegative variables λ, w is finite precisely when

$$
\sum_y \gamma_{i,y}^{(k)} = \alpha / n_k, \qquad \alpha + \mu^\top z = 1,
$$

$$
\sum_m \beta_{km} \le \rho_k \alpha, \qquad \beta_{km} \ge c_k(\gamma)[m] - \rho_k \mu_m, \qquad \beta_{km} \ge 0.
$$

Summing the row equations within each class gives

$$
\sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = \sum_{i=1}^{n_k} \frac{\alpha}{n_k} = \alpha, \qquad k = 0,1.
$$

Averaging these two equalities yields $M(\gamma) = \alpha$. For each k, m, the smallest admissible multiplier is $\beta_{km} = (c_k(\gamma)[m] - \rho_k \mu_m)_+$. Thus the constraints on β are feasible if and only if

$$\sum_{m=1}^{D} \big(c_k(\gamma)[m] - \rho_k \mu_m\big)_+ \le \rho_k M(\gamma), \qquad k = 0, 1.$$

Thus eliminating α and β leaves precisely $(\gamma, \mu) \in \mathcal{D}$ and $M(\gamma) + \mu^\top z = 1$.

The empty-selection epigraph is feasible in Eq. (11) for every $z \ge 0$ with $w = 0$; the diagonal constraints bound its objective below. Its finite sublevels are bounded and closed as in Appendix A. For any feasible tuple, the replacements $a_i^{(k)} \mapsto a_i^{(k)} + \delta$ and $\eta \mapsto \eta + 3\delta$, with $\delta > 0$, make every loss inequality strict while preserving all affine constraints. Slater's condition with affine constraints therefore yields

$$v(z) = \max_{\substack{(\gamma,\mu)\in\mathcal{D} \\ M(\gamma)+\mu^\top z = 1}} H(\gamma). \tag{12}$$

It remains to remove the normalization. The constraints defining $\mathcal{D}$ are homogeneous in (γ, μ), and $H(s\gamma) = sH(\gamma)$ for $s \ge 0$. If $M(\gamma) + \mu^\top z = 0$, the row constraints force $\gamma = 0$, and the objective in Eq. (5) is zero. Otherwise, division by $M(\gamma) + \mu^\top z$ produces a feasible pair in Eq. (12). Since

$$2\sqrt{sh} - s = h - (\sqrt{s} - \sqrt{h})^2, \qquad \max_{s\ge 0}\{2\sqrt{sh} - s\} = h \quad (h \ge 0), \tag{13}$$

we obtain

$$\begin{aligned} F(z) &= \max_{\substack{(\gamma,\mu)\in\mathcal{D} \\ M(\gamma)+\mu^\top z = 1}} \ \max_{s\ge 0}\{2\sqrt{sH(\gamma)} - s\} \\ &= \max_{\substack{(\gamma,\mu)\in\mathcal{D} \\ M(\gamma)+\mu^\top z = 1}} H(\gamma) = v(z). \end{aligned}$$

The scalar maximum is attained at $s = h$, so attainment in Eq. (12) also gives attainment in Eq. (5). In particular, $F(\mathbf{1}_S) = V(S)$.

Finally, $\mathcal{D}$ is convex, $\sqrt{H}$ is concave, and M is linear. Hence Eq. (5) is a convex optimization problem for fixed z. Its objective is affine in z and its feasible set is independent of z, so F is convex. The correspondence $S \leftrightarrow \mathbf{1}_S$ gives Eq. (6). □

References

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